



SCOPE OF ACCREDITATION TO ISO/IEC 17025:2005

ASSOCIATED ENVIRONMENTAL SYSTEMS

8 Post Office Square

Acton, MA 01720

Ryan Miller Phone: 978 772 0022

CALIBRATION

Valid To: July 31, 2018

Certificate Number: 3964.01

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory to perform the following calibrations¹:

I. Electrical – DC/Low Frequency

Parameter/Equipment	Range	CMC ² (±)	Comments
DC Voltage – Generate ³ Fixed Points	110 mV 1 V 15 V	0.8 mV 0.02 V 0.08 V	Fluke 753
DC Resistance – Generate ³	(0 to 10) Ω (0 to 100) Ω (0 to 1000) Ω	0.007 Ω 0.09 Ω 0.7 Ω	Fluke 753
DC Current – Generate ³	(0 to 22) mA	0.12 mA	Fluke 753
Electrical Calibration of Thermocouple Indicating Systems ³ Type T Type J Type K	 (-200 to 0) °C (0 to 400) °C (0 to 1200) °C (0 to 1372) °C	 0.68 °C 1.2 °C 0.79 °C 0.93 °C	 Fluke 753
Electrical Calibration of RTD Indicating Systems ³ Pt 385 100 Ω	 (200 to 250) °C	 0.72 °C	 Fluke 753

II. Thermodynamics

Parameter/Equipment	Range	CMC ² ($\pm$)	Comments
Relative Humidity – Measure ³	(10 to 95) % RH	3.6 % RH	Vaisala HMP 77B
Temperature – Measure ³	(-80 to 0) °C (0 to 200) °C (0 to 760) °C (0 to 1100) °C	1.4 °C 2.5 °C 2.3 °C 1.7 °C	Fluke 753 with T, J and K thermocouples

¹ This laboratory offers commercial and field calibration service.

² Calibration and Measurement Capability Uncertainty (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. CMCs represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of $k = 2$. The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

³ Field calibration service is available for this calibration and this laboratory meets A2LA R104 – *General Requirements: Accreditation of Field Testing and Field Calibration Laboratories* for these calibrations. Please note the actual measurement uncertainties achievable on a customer's site can normally be expected to be larger than the CMC found on the A2LA Scope. Allowance must be made for aspects such as the environment at the place of calibration and for other possible adverse effects such as those caused by transportation of the calibration equipment. The usual allowance for the actual uncertainty introduced by the item being calibrated, (e.g. resolution) must also be considered and this, on its own, could result in the actual measurement uncertainty achievable on a customer's site being larger than the CMC.



Accredited Laboratory

A2LA has accredited

ASSOCIATED ENVIRONMENTAL SYSTEMS

Acton, MA

for technical competence in the field of

Calibration

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2005 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets any additional program requirements in the field of calibration. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (refer to joint ISO-ILAC-IAF Communiqué dated 8 January 2009).

Presented this 13th day of April 2016.



A handwritten signature in blue ink, appearing to read 'J. C. Bunt'.

Senior Director of Quality and Communications
For the Accreditation Council
Certificate Number 3964.01
Valid to July 31, 2018

For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.